CyberBOT: Ontology-Grounded Retrieval Augmented Generation for Reliable Cybersecurity Education

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Abstract

Advancements in large language models (LLMs) have enabled the development of intelligent educational tools that support inquiry-based learning across technical domains. In cybersecurity education, where accuracy and safety are paramount, systems must go beyond surface-level relevance to provide information that is both trustworthy and domain-appropriate. To address this challenge, we introduce CyberBOT[1], a question-answering chatbot that leverages a retrieval-augmented generation (RAG) pipeline to incorporate contextual information from course-specific materials and validate responses using a domain-specific cybersecurity ontology. The ontology serves as a structured reasoning layer that constrains and verifies LLM-generated answers, reducing the risk of misleading or unsafe guidance. CyberBOT has been deployed in a large graduate-level course at Arizona State University (ASU),[2] which illustrates a promising direction for developing reliable and curriculum-aligned AI applications in specialized educational contexts.

CCS Concepts

• **Computing methodologies** → **Natural language generation**.

Keywords

Question Answering, Retrieval Augmented Generation, Cybersecurity Education, Large Language Model

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1 Introduction

The integration of large language models (LLMs) into educational applications has introduced new opportunities for personalized and interactive learning experiences [4, 13, 15]. In particular, question-answering (QA) systems powered by LLMs offer the potential to support self-paced inquiry and deepen conceptual understanding [6, 17]. However, despite their generative capabilities, LLMs often suffer from factual inaccuracies and hallucinations [7], mainly when applied to high-stakes and technically demanding domains

[1]Code: https://github.com/rccrdmr/CyberBOT
[2]Video: https://youtu.be/X2WorBxOQHo

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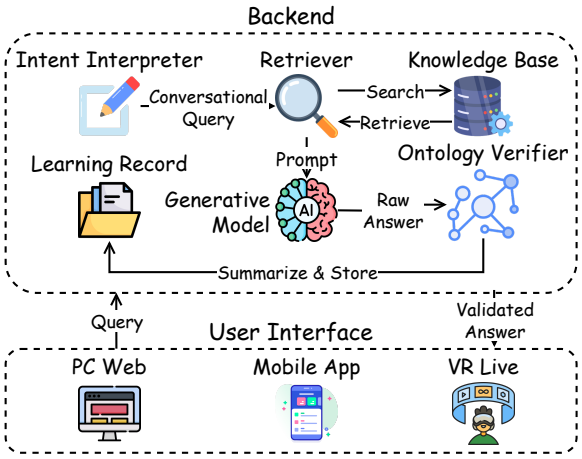


Figure 1: Framework of proposed CyberBOT. Students submit queries to UI and get responses from the backend.

such as cybersecurity education [12]. The risks associated with such outputs, ranging from conceptual misunderstanding to propagation of unsafe practices [10, 11, 18], underscore the need for enhanced reliability in educational settings.

Retrieval-augmented generation (RAG) has emerged as a common strategy to improve response reliability by conditioning model outputs on retrieved external documents [8]. Although this approach increases contextual relevance, it does not offer a guarantee of correctness, particularly when the retrieved context is ambiguous or incomplete [3, 16]. Consequently, RAG-based systems remain susceptible to producing only loosely grounded responses in the underlying knowledge base, leading to challenges in ensuring content validity and safety in the education landscape.

To address these limitations, we propose CyberBOT, an ontology-grounded RAG system tailored for reliable cybersecurity education. CyberBOT consists of three sequential stages, each contributing to the reliability and contextual relevance of the system's output as illustrated in Figure 1. First, an intent interpreter analyzes the multi-turn conversational history to infer the student's underlying intent. This component reformulates the user query into a context-enriched, knowledge-intensive version, thereby enabling more effective retrieval in multi-round interactions. Second, based on the interpreted intent, relevant documents are retrieved from a curated course-specific knowledge base using retrieval-augmented generation techniques. These documents serve as the contextual foundation for generating an initial response with the LLM. Finally, the generated answer is validated using a domain-specific cybersecurity ontology, which ensures semantic alignment with authoritative knowledge and filters out hallucinated or unsafe content. This three-stage architecture allows CyberBOT to address both contextual ambiguity and factual correctness, significantly enhancing the trustworthiness and educational value of the system in real-world instructional settings.

A key strength of CyberBOT lies in its real-world deployment. The system has been integrated into a live classroom setting and is accessible to more than one hundred graduate students enrolled in the spring 2025 semester of CSE 546: Cloud Computing course at Arizona State University (ASU). Students interact with the system via a dedicated web interface, submitting course-related queries and receiving validated responses to assist with their studies. This deployment enables direct observation of system usage in an authentic educational environment and provides a valuable opportunity to evaluate the practical utility and pedagogical effectiveness of ontology-informed validation in question-answering systems. Overall, CyberBOT illustrates a promising direction for developing reliable and curriculum-aligned AI applications in specialized educational contexts. Our contributions are summarized as:

- ⋆ We propose a novel QA system CyberBOT that combines RAG with cybersecurity ontologies for answer validation, improving factual accuracy in a specialized domain.
- ⋆ We construct a knowledge base from class materials to ground the QA system's responses in relevant, contextually accurate information aligned with the course curriculum.
- ⋆ We deploy CyberBOT as a user-friendly Q&A platform, in a live classroom with more than one hundred graduate students, with the goal of providing practical insights into the system's performance and student engagement.

2 The proposed CyberBOT

We introduce CyberBOT, an ontology-aware RAG system designed for multi-turn QA in cybersecurity education, as illustrated in Figure 1. A step-by-step example with data flow is shown in Figure 2.

2.1 Intent Interpreter

Given a domain-specific question q, we leverage an intent model $\mathcal{I}$ as an intent interpreter to capture the user's intention from the last k-round history conservation c. Then, the intent model will rewrite the current query as a knowledge-intensive conversation query q_c: $q_c = \mathcal{I}(q, c)$, which not only enriches the context for the generation and enables multi-turn retrieval. In the implementation, we design a semantic rule-based classifier to determine if a question needs to be written or not to reduce the computational cost.

2.2 Retrieval Augmented Generation

Based on the conversation query, the model retrieves the related document from a knowledge base. Then, the augmented context is used to prompt LLMs to generate the answer.

2.2.1 Knowledge Base. The knowledge base plays a critical role in supporting the responses of CyberBOT and consists of two main components: (I) A collection of common cybersecurity QA pairs curated by domain experts, derived from laboratory instruction manuals used in graduate-level advanced cybersecurity courses. These cover topics such as building intrusion detection systems and monitoring system activity. (II) Course materials from CSE 546: Cloud Computing at ASU, including lecture slides, assignments, quizzes, and project instructions. Most of these resources are in PDF format, which we preprocess into smaller, semantically meaningful chunks before storing them in the knowledge base.

2.2.2 Retriever. For each turn in the conversation, given the conversation query q_c, the retriever module $\mathcal{R}$ selects the most relevant document d from the corresponding course-specific knowledge base by computing similarity: $d = \mathcal{R}(q_c, kb)$. To accelerate retrieval, all documents are pre-encoded into vector representations, allowing for efficient similarity search within the knowledge base.

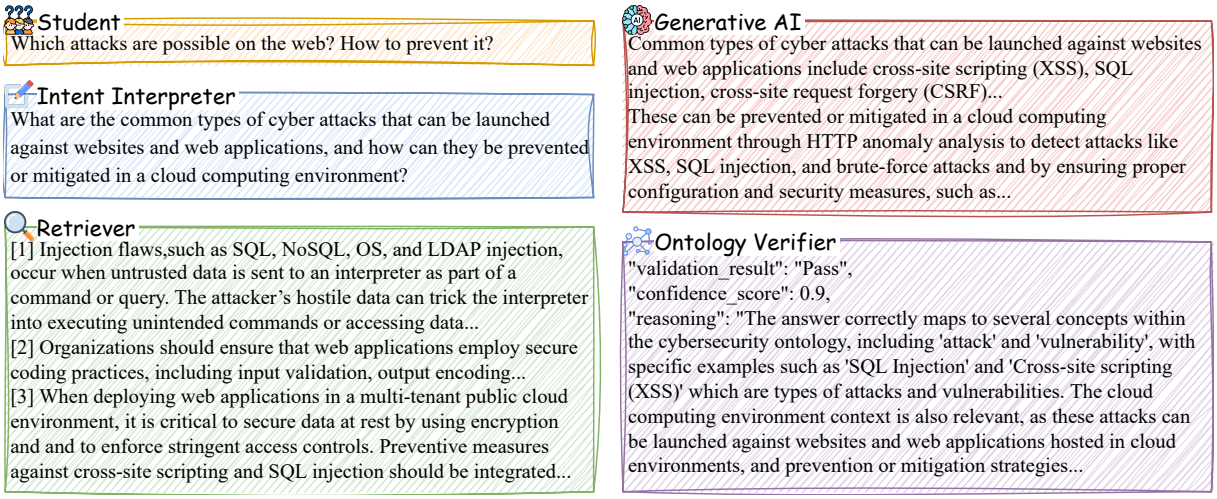


Figure 2: Illustrative of data flow in CyberBOT. The response is augmented and validated in various flows.

2.2.3 Generation. The user query and related document are used to prompt LLM $\mathcal{G}$ to generate a preliminary answer: $a = \mathcal{G}(q_c, d)$.

2.3 Ontology-based Validation

In practice, validating the generated answers is essential to prevent misinformation or misuse. To enable this, we design an ontology-based validation mechanism grounded in domain knowledge. As described earlier, a knowledge graph (KG) captures domain-specific factual triples in the form of entity-relationship-entity; on top of that, an ontology defines high-level domain concepts and their semantic relationships in a structured hierarchy.

Our validation process begins by extracting and analyzing key concepts and their relationships from the course materials. These are then distilled by cybersecurity experts into a domain-specific ontology o, encapsulating essential patterns and logical structures. Finally, we employ an ontology verifier $\mathcal{V}$ to assess whether the generated answer aligns with the ontology: $r = \mathcal{O}(q_c, a, o)$, where $r \in (0, 1)$ represents the validation score. Answers falling below a certain threshold are flagged and rejected to ensure the reliability and domain-appropriateness of the system's responses.

2.4 Student Learning History

To facilitate personalized learning, we designed a user management system, which tracks each user's learning log and stores it in the backend while anonymizing personal information.

3 System Deployment

Our system architecture, as shown in Figure 3, is designed to streamline the entire workflow from data ingestion to end-user interaction and evaluation. Here, we discuss the deployment stage, detailing the user-facing and backend components.

3.1 User Interface

We build a simple, web-based front-end using Streamlit, an open-source Python framework that allows rapid development of interactive web apps. Users enter queries or select tasks, and the interface displays the system's responses in real-time.

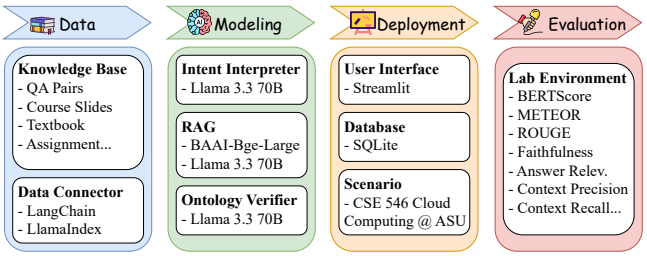


Figure 3: Pipeline and implementation details of the system.

3.2 Backend

All domain-specific materials (e.g., QA pairs, slides, assignments) are stored in a unified repository. Texts are split into chunks with 512 tokens to facilitate efficient retrieval. Each chunk is embedded using BAAI-Bge-Large-1.5 and stored in a FAISS index. User queries are similarly embedded, and nearest-neighbor search identifies the most relevant chunks. A Llama 3.3 70B model classifies queries according to task-specific intentions. Based on these results, the system retrieves the top-3 relevant chunks from FAISS for context. Another Llama 3.3 70B model fuses the retrieved context and the user query to generate a coherent response. This step handles reasoning and synthesizes domain knowledge for the final output. The Ontology Verifier, powered by Llama 3.3 70B, evaluates the generated responses by verifying their alignment with our domain-specific ontology. A lightweight SQLite database is used to store session data and interaction logs for subsequent analysis.

3.3 Device and Hardware

We encapsulate the CyberBOT into a Docker environment and develop the system in a dedicated server with an A100 80GB. For convenience, we use Together AI API to enable all the embedding models or LLMs. For security reasons, a VPN with ASU credentials will be needed to access the system outside the university network.

4 Application

We have deployed CyberBOT into CSE 546: Cloud Computing for the 2025 Spring semester. A live example is shown in Figure 4.

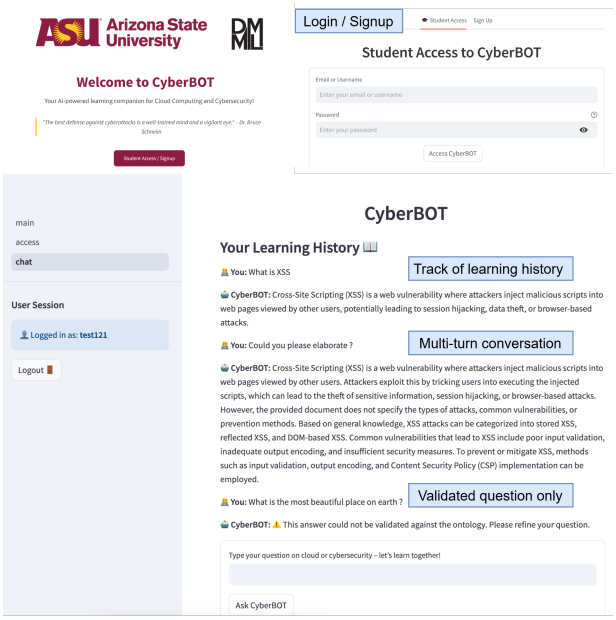


Figure 4: A live running example of CyberBOT system.

4.1 Use-Case Illustration

Users can input domain-specific questions through the dialogue interface. The RAG system retrieves relevant documents from the appropriate knowledge base and generates ontology-validated responses. If a query falls outside the scope of the defined domain ontology, the response is rejected to maintain reliability and relevance. Given the highly specialized nature of the knowledge base, the system serves as an effective AI assistant for: (I) learning cybersecurity and cloud computing concepts grounded in the course material, (II) assisting with assignment-related queries by providing step-by-step explanations, and (III) offering fine-grained, hands-on guidance for course projects, such as code completion and framework design. Additionally, all interactions, including user queries and corresponding responses, are stored under each user's account. These historical conversation logs can serve as a foundation for developing individualized learning paths and enabling personalized educational experiences in the future.

4.2 Computational Evaluation

Dataset. To evaluate the performance of CyberBOT in a controlled lab setting, we use CyberQ [1], an open-source dataset comprising approx 3,500 open-ended cybersecurity QA pairs across topics such as tool usage, setup instructions, attack analysis, and defense techniques. The dataset includes questions of varying complexity, categorized into ZS (1,027), FS (332), and OD (2,171) types, making it well-suited for testing multi-level question answering.

Metric. We consider two categories of metrics: (I) QA-based metrics, which evaluate the quality of generated answers, including BERTScore [14], METEOR [2], ROUGE-1 [9], and ROUGE-2 [9]. (II) RAG-based metrics, which assess the retrieval effectiveness and accuracy covering Faithfulness, Answer Relevancy, Context Precision, Context Recall, and Context Entity Recall. We implement the RAG-based metric using RAGAS [5].

Metric	ZS	FS	OD	AVG
QA-based Metrics				
BERTScore ↑	0.929	0.946	0.933	0.933
METEOR ↑	0.786	0.859	0.786	0.793
ROUGE-1 ↑	0.649	0.788	0.641	0.657
ROUGE-2 ↑	0.598	0.720	0.593	0.606
RAG-based Metrics				
Faithfulness ↑	0.813	0.891	0.760	0.788
Answer Relevancy ↑	0.983	0.986	0.983	0.983
Context Precision ↑	0.989	1.000	0.996	0.994
Context Recall ↑	0.991	0.997	0.995	0.994
Context Entity Recall ↑	0.939	0.951	0.967	0.957

Table 1: Performance of CyberBOT across various datasets.

Main result. We summarize the main result in Table 1. (I) Generally, the proposed tool achieves satisfaction across both QA-based metrics and RAG-based metrics, e.g., CyberBOT achieves an average of BERTScore and Context Recall of 0.933 and 0.994, respectively, which indicates the system not only generates high-quality answers but also can retrieve the very relevant document from the knowledge base. (II) The framework produces higher scores in the FS category than those in ZS and OD, which may be because the QA pairs in the FS leverage in-context learning examples and are thus more consistent. (III) Among the QA-based metrics, the system achieves superior performance under BERTScore compared to others. It suggests that the generated answer has good overall semantic similarity while producing various words and paraphrases. (IV) Among the RAG-based metrics, we can observe that the results under Answer Relevancy, Context Precision, and Context Recall are very competitive, which showcases that the framework benefits from related documents as references. However, the Faithfulness score is slightly lower; one possible explanation is that the model will leverage its own knowledge to generate answers when there is no closely relevant material in the knowledge base.

5 Conclusion

We present CyberBOT, an ontology-grounded RAG assistant designed to support reliable and context-aware cybersecurity education. CyberBOT leverages an intent interpreter to capture the educational dialogue context and employs an ontology verifier to ground generated responses in domain-specific rules and constraints. The system has been deployed in CSE 546: Cloud Computing, serving over one hundred graduate students. This work not only demonstrates the potential of specialized natural language processing systems in education but also opens new avenues for advancing ontology-guided approaches in cybersecurity learning. Future work will focus on enhancing the system with personalized learning capabilities tailored to individual users.

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GenAI Usage Disclosure

To enhance clarity and readability, we utilized OpenAI GPT-4.5 exclusively as a language polishing tool. Its role was confined to proofreading, grammatical correction, and stylistic refinement—functions analogous to those provided by traditional grammar checkers and dictionaries. This tool did not contribute to the generation of new scientific content or ideas, and its usage is consistent with standard practices for manuscript preparation.

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